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Applications of meromorphic functions in circuit theory and dynamical systems analysis

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Abstract

The relationship between RLC circuits and complex analysis is profound, as the description and modelling of the behaviour of these circuits involves the use of complex numbers to represent impedances, currents, and voltages. Complex functions play a central role, and in the case of RLC circuits, they can be extended to the analysis of meromorphic functions, which help describe critical points such as resonance and attenuation.

In the first part of the paper, meromorphic functions are defined on both the complex and extended planes, with important properties described that are useful for interpreting singularities. The second part revolves around concepts related to dynamic systems, stability criteria, and transfer functions, aiming to model resonance in an RLC circuit. The final section is dedicated to studying the variation of circuit parameters, highlighting the connection between the poles of meromorphic functions and the fundamental characteristics of the circuit, such as resonance, response frequency, and quality factor.

The entire study highlights certain properties of poles and explains specific behaviours of resonance, response frequency, and quality factor, proving useful in the design of electrical circuits. For example, a useful conclusion is that a sharp peak occurs at the resonance frequency for the impedance of the parallel combination.

The paper includes numerical simulations in MATLAB, underlining how the position of poles and zeros influences the stability and performance of systems. The obtained results confirm the usefulness and elegance of the meromorphic formalism in addressing complex engineering problems.

1. Introduction

Complex analysis, as a central branch of modern mathematics, plays an essential role in describing and interpreting complex physical phenomena, providing a robust theoretical framework for the study of dynamical systems and electrical circuits. Through holomorphic and meromorphic functions, which possess remarkable analytical properties and a well-defined structure of singularities, it becomes possible to mathematically model fundamental physical processes, such as signal propagation or the frequency behavior of linear systems. Closely linked to physics, complex analysis allows the transcription of engineering problems into mathematical language, thus facilitating the approach to aspects such as stability, resonance, or system performance.

In the field of electrical circuits, meromorphic functions naturally arise in the analysis of transfer functions, where the distribution of poles and zeros determines the essential characteristics of the system's response. For example, the impedance and admittance functions of RLC circuits can be expressed as rational functions, and their study in the complex plane provides critical information about stability and signal dynamics. Similarly, in the theory of dynamical systems, representation through meromorphic functions allows the investigation of causality properties and the system's response to various inputs.

2. Meromorphic Functions

2.1 Definition

A meromorphic function on an open set D in the complex plane is a function that is holomorphic on all of D, except for a set of isolated points, $p_1, p_2, \cdots$, which are the poles of the function. Denoting $\mathrm{D}^* = \mathrm{D} \backslash \{p_1, p_2, \cdots\}$, we have:

- $f: \mathrm{D}^* \to \mathbb{C}$ is holomorphic.
- $p_1, p_2, \cdots$ are the poles for f.

Consider a rational function on $\mathbb{C}$ of the form $R(z) = \frac{P(z)}{Q(z)}$, where $P(z)$ and $Q(z)$ are polynomials.

2.2 Singularities of Meromorphic Functions

Definition 1 Let E be an open set in $\mathbb{C}$ and $f: E \to \mathbb{C}$ be a given function. A point $z_0 \in \mathbb{C}$ is called an ordinary point for f if there exists an open disk centered at z_0, $\Delta(z_0, r) = \{z \in \mathbb{C} | |z - z_0| < r\}$, on which f is holomorphic. A point $z_0 \in \mathbb{C}$ is called a singular point for f if z_0 is not an ordinary point for f, i.e., f is not holomorphic on any disk $\Delta(z_0, r)$.

Definition 2 Let $f: D \to \mathbb{C}$ and let $z_0 \in \mathbb{C}$ be a singular point for f, an accumulation point for D.

- z_0 is called a **removable singularity** if $\lim_{z \to z_0} f(z) \in \mathbb{C}$.
- z_0 is called a **pole of order k** $\in \mathbb{N}^*$ for f if the function $g(z) = (z - z_0)^k \cdot f(z)$ has an ordinary point at z_0. This is equivalent to the fact that

$$\lim_{z \to z_0} f(z) = \infty \quad \text{and} \quad \lim_{z \to z_0} (z - z_0)^k \cdot f(z) \in \mathbb{C}^*.$$

The order of the pole is the smallest non-zero natural number k for which the preceding limit is finite and non-zero. In the case $k = 1$, we say that z_0 is a simple

pole or a pole of order 1. If $k = 2$, then z_0 is a double pole or a pole of order 2, and if $k = 3$, then we say that z_0 is a triple pole or a pole of order 3.

- z_0 is called an essential singular point if it is neither removable nor a pole, i.e.,

$$\lim_{z \to z_0} f(z) \quad \text{does not exist(or is infinite, but not a pole).}$$

2.3 Meromorphic Functions on the Extended Plane

Another important characteristic of meromorphic functions is their connection to the extended plane or the Riemann sphere, which is a compactification of the complex plane. On the Riemann sphere, we add a point "at infinity" (∞) to the complex plane, thus obtaining a compact structure. Thus, meromorphic functions are not only defined on the entire complex plane, except for isolated singularities, but their behavior at infinity (i.e., near the point ∞) can also be studied in a more general mathematical framework. We consider the point at infinity, $z = \infty$, to have the same importance as any other point, considering the extended complex plane: $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$.

Thus, meromorphic functions are functions $f: D \backslash \{p_1, \dots, p_j, \dots\} \to \mathbb{C}$, for which the extended function $\hat{f}: D \to \hat{\mathbb{C}}$, is defined by $\hat{f}(p_j) = \infty$ for all poles p_j.

For a meromorphic function on $\mathbb{C}$, extended to $\hat{\mathbb{C}}$, we say that $z = \infty$ is a pole / removable singularity / essential singularity if $z = 0$ is a pole / removable singularity / essential singularity for the function $\hat{f}(z) = f\left(\frac{1}{z}\right)$. We say that a meromorphic function on $\mathbb{C}$ is meromorphic on the extended plane if $z = \infty$ is not an essential singularity.

Definition 3. Let $D \subseteq \mathbb{C}$ be an open set, $f \in M(D)$ and $a \in D$. If f is not identically zero, and its Laurent series centered at point a has coefficients c_n, $n \in \mathbb{Z}$, the number

$$\operatorname{ord}_a(f) := \inf\{n \in \mathbb{Z} | c_n \neq 0\}.$$

is called the **order of the function f at point a.**

If f_e is the extension with values in $\mathbb{C} \cup \{\infty\}$ of the function f defined above, and $f_e(a) = 0$, we say that the point a is a zero of order $\operatorname{ord}_a(f)$ of the function f. The point a is called a simple zero of f if $\operatorname{ord}_a(f) = 1$.

If $a \in P(f; D)$ (the set of poles of f in D), then $-\operatorname{ord}_a(f)$ is called the order of the pole a (Note: original had $\operatorname{ord}_a(f) = -\operatorname{ord}_a(f)$, corrected to define order of pole).

For the identically zero function 0_D, we define $\operatorname{ord}_a(0_D) = \infty$, $\forall a \in D$.

Proposition 1: Let $D \subseteq \mathbb{C}$ be a non-empty open set, and $f \in M(D)\backslash\{0_D\}$ be a non-identically zero meromorphic function on D. For $a \in D$, the following properties hold:

- a is a regular point for f if and only if $\mathrm{ord}_a(f) \geq 0$.
- a is a regular point for f with $f_e(a) \neq 0$ if and only if $\mathrm{ord}_a(f) = 0$.
- a is a pole for f if and only if $\mathrm{ord}_a(f) < 0$.

If also, $g \in M(D)\backslash\{0_D\}$ is non-identically zero, and $\lambda \in \mathbb{C}\backslash\{0\}$, then

- $\mathrm{ord}_a(f \cdot g) = \mathrm{ord}_a(f) + \mathrm{ord}_a(g)$.
- $\mathrm{ord}_a(f + g) \geq \min(\mathrm{ord}_a(f), \mathrm{ord}_a(g))$, with equality if and only if $\mathrm{ord}_a(f) \neq \mathrm{ord}_a(g)$.

Proposition 2: The set $M(\mathbb{C} \cup \{\infty\})$ of meromorphic functions on $\mathbb{C}$ for which ∞ is not an essential singularity coincides with the set of rational functions.

Lemma 1: A rational function admits a pole or a removable singularity at infinity. The singularity at infinity is removable if and only if $\deg Q \geq \deg P$.

Theorem 1: The only meromorphic functions on $\hat{\mathbb{C}}$ are the rational functions.

Theorem 2 (Argument Principle Application): Let $w = w(t)$ be a meromorphic function in a simply connected domain D, and γ be a simple closed continuous curve in D that does not pass through any of the poles or zeros of the function. Then:

$$\frac{1}{2\pi i}\int_\gamma \frac{w'(z)}{w(z)}\, dz = N - P$$

where:

- N is the number of zeros,
- P is the number of poles inside the curve γ,

counting multiplicities.

Definition 4: $\oint_\gamma d(Ln(w(t))) = \Delta_\gamma Ln(w(z))$ represents the variation of the function $Ln(w(z))$ upon traversing the curve γ in the positive direction.

Theorem (Argument Principle): Under the conditions of Theorem 2, we have:

$$\boldsymbol{\frac{1}{2\pi}\Delta_\gamma \mathrm{arg} w(z) = N - P}$$

Theorem 3 (Rouché's Theorem): Let $f(z)$ and $g(z)$ be two functions holomorphic in a simply connected domain containing the simple closed curve γ, such that:

$$|f(z)| > |g(z)|, \quad \text{forall} z \in \gamma$$

Then the functions $f(z)$ and $F(z) = f(z) + g(z)$ have the same number of zeros in the domain D bounded by the curve γ.

3. Importance of Meromorphic Functions in Physics

Meromorphic functions hold a central position in the mathematical modeling of various physical phenomena due to their ability to describe analytical behaviors with well-defined singularities. In particular, these functions appear in the study of wave propagation, quantum field theory, and dynamical systems, where their structure of poles and zeros reflects essential physical properties, such as resonances or stability. For example, in scattering theory, meromorphic functions allow the representation of scattering amplitudes through analytical properties that encode information about bound states and the continuum.

In quantum mechanics, the S-matrix (scattering matrix) is often analyzed as a meromorphic function of energy, with poles corresponding to resonant states. Similarly, in electrical circuit theory, transfer functions can be expressed as ratios of polynomials, where the distribution of poles in the complex plane determines the frequency response of the system. Thus, meromorphic functions not only provide a rigorous mathematical framework for addressing physical problems but also facilitate deep connections between systems theory, spectral analysis, and wave mechanics.

4. Study of the Stability of Dynamical Systems

4.1. Stability of Dynamical Systems

We consider linear time-invariant (LTI) dynamical systems with a single input and a single output. A dynamical system is stable if every bounded excitation corresponds to a bounded response. The relationship between the excitation $x(t)$ and the system response $y(t)$ is mathematically modeled by a linear differential equation with constant (real) coefficients of the form:

$$a_0 \frac{d^n y}{dt^n} + a_1 \frac{d^{n-1} y}{dt^{n-1}} + \cdots + a_{n-1} \frac{dy}{dt} + a_n y = x(t)$$

The general solution to this equation has the form:

$$y(t) = y_p(t) + c_1 e^{s_1 t} + \cdots + c_n e^{s_n t}$$

where $y_p(t)$ is the particular solution of the non-homogeneous equation, $c_1, c_2, \dots, c_n$ are arbitrary constants (resulting from the initial conditions of the problem), and $s_1, s_2, \dots, s_n$ are the roots of the characteristic equation

$$A(s) \equiv a_0 s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n = 0.$$

Remark:

• In the theory of dynamical systems, the complex variable is often denoted as $s = \sigma + i\omega$ (using sigma instead of r).

• If all roots $s_k = \sigma_k + i\omega_k$ lie in the half-plane $\mathrm{Re}\{s\} < 0$ (the left half-plane), then the system is stable. For $\mathrm{Re}\{s\} > 0$, the system is unstable.

If the characteristic equation has simple roots with zero real part, it is necessary to also consider the behavior of the function $x(t)$ to analyze the system's stability (marginal stability).

4.2 Stability Criteria

Stodola's Condition:

A necessary condition for all roots of the characteristic equation to have negative real parts is that all coefficients of the equation have the same sign (assuming $a_0 > 0$).

Hurwitz Criterion:

For all roots of the polynomial (3) with real coefficients a_k ($a_0 > 0$) to have negative real parts, it is necessary and sufficient that the following inequalities are met

$$D_1 = a_1 > 0, \quad D_2 = \begin{vmatrix} a_1 & a_0 \\ a_3 & a_2 \end{vmatrix} > 0, \quad D_3 = \begin{vmatrix} a_1 & a_0 & 0 \\ a_3 & a_2 & a_1 \\ a_5 & a_4 & a_3 \end{vmatrix} > 0,$$

$$\vdots$$

$$D_n = \begin{vmatrix} a_1 & a_0 & 0 & \cdots & 0 \\ a_3 & a_2 & a_1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{2n-1} & a_{2n-2} & a_{2n-3} & \cdots & a_n \end{vmatrix} > 0,$$

where $a_k = 0$ for $k > n$.

Nyquist Criterion:

For the dynamical system described by the meromorphic function $A(s)$ (often the open-loop transfer function, or related characteristic function), having P poles in the right half-plane $\mathrm{Re}\{s\} > 0$, to be stable (in closed loop), it is necessary and sufficient that as s traverses the Nyquist contour γ (enclosing the RHP), its image $w = A(s)$ encircles the origin P times in the clockwise direction. (Note: Stability often refers to the closed-loop system whose characteristic equation involves A(s), e.g., 1+A(s)=0, making the critical point -1).

Nyquist Criterion for Feedback Systems:

For the automatic control system, having the closed-loop transfer function:

$$H(s) = \frac{kG(s)}{1+kG(s)} \quad \Rightarrow \quad \text{Characteristic Eq: } 1 + kG(s) = 0, \text{or} G(s) = -1/k$$

to be stable, it is necessary and sufficient that, as the point s traverses the Nyquist contour γ in the positive (counter-clockwise) direction, the locus of $G(s)$ encircles the point $-1/k$ P times in the counter-clockwise direction, where P is the number of poles of the open-loop function $G(s)$ in the right half-plane $\text{Re}\{s\} > 0$. (Adjusted the wording and direction for standard stability criterion based on encirclements of -1/k).

5. The RLC Circuit

An RLC circuit is an electrical circuit consisting of three components: an inductor (L), which stores energy in a magnetic field; a resistor (R), which opposes the flow of current and dissipates energy as heat; and a capacitor (C), which stores energy in an electric field. These components can be connected in series or in parallel in an alternating current (AC) circuit. Together, they influence the overall behavior of the circuit by controlling the current flow, leading to a phenomenon called resonance at a specific frequency, where energy transfer is maximized.

5.1. The RLC Circuit.

In an RLC circuit, the behavior of **current and voltage** across each component is different due to the specific interaction of each element with the alternating current.

The resistor obeys Ohm's Law, expressed as:

$$V_R = I \cdot R \tag{1}$$

where:
- V_R is the voltage across the resistor;
- I is the current flowing through the circuit;
- R is the electrical resistance.

Voltage and current are in phase, meaning they reach their maximum and minimum values simultaneously. There is no delay between the voltage and current waveforms.

The inductor stores energy in the form of a magnetic field when current passes through it. The voltage across an inductor depends on the rate at which the current changes. The inductor opposes the current flow through a physical quantity called inductive reactance, defined by:

$$X_L = \omega L = 2\pi f L \tag{2}$$

where: • X_L is the inductive reactance;

• f is the frequency of the AC signal in Hertz;

• L is the inductance.

The voltage across the inductor is:

$$V_L = IX_L \tag{3}$$

In an inductor, the voltage leads the current by 90 degrees. This means the voltage reaches its maximum value before the current does. This phenomenon occurs because the inductor opposes changes in current, requiring time to build up or dissipate magnetic energy.

The capacitor stores energy in an electric field between its plates. The voltage across the capacitor is related to the amount of accumulated charge. Capacitive reactance is analogous to the resistance of a resistor and measures the opposition to current:

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C} \tag{4}$$

where: • X_C is the capacitive reactance;

• C is the electrical capacitance.

The voltage is expressed by:

$$V_C = IX_C \tag{5}$$

In a capacitor, the current leads the voltage by 90 degrees, opposite to the inductor. This means the current reaches its maximum value before the voltage reaches its maximum value, as the capacitor needs to charge progressively.

The total current in an RLC circuit depends on the combined effects of the resistor's resistance, the inductor's inductive reactance, and the capacitor's capacitive reactance. The **impedance** Z of the circuit determines the amount of current flowing for a given voltage $V(t)$ (Changed U to V):

$$Z = \sqrt{R^2 + (X_L - X_C)^2} \tag{6}$$

Thus, the current $I(t)$ is:

$$I(t) = \frac{V(t)}{Z} \tag{7}$$

In an RLC circuit, **resonance** occurs when the inductive reactance and capacitive reactance cancel each other out, allowing the circuit to transfer energy maximally. Using the formulas for inductive and capacitive reactances:

$$X_L = \omega L \quad \text{and} \quad X_C = \frac{1}{\omega C} \tag{8}$$

At resonance, these become equal:

$$\omega L = \frac{1}{\omega C} \tag{9}$$

Solving for the resonant angular frequency ω_0, we get:

$$\omega_0 = \frac{1}{\sqrt{LC}} \tag{10}$$

The resonant frequency (in Hz) is given by:

$$f_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi\sqrt{LC}} \tag{11}$$

This frequency is essential in applications such as radio tuning, where the circuit selects a specific signal frequency, filtering out others. At resonance, the circuit's impedance is minimal and equals only the resistance:

$$Z = \sqrt{R^2 + (X_L - X_C)^2} \Rightarrow Z_{min} = R \quad (\text{when} X_L = X_C) \tag{12}$$

Since the impedance is minimized, the current in the circuit is maximized, according to Ohm's Law:

$$I_{max} = \frac{V}{Z_{min}} = \frac{V}{R} \tag{13}$$

At resonance, the circuit allows maximum current flow for a given voltage. This phenomenon is essential in applications involving filtering and amplifying signals at precise frequencies.

In an RLC circuit, **damping** represents the reduction in the amplitude of oscillations due to energy dissipation in the resistive element. This phenomenon is described by the quality factor Q and the damping constant α, which characterize the rate of decrease of the oscillation amplitude. For a series RLC circuit, the damping constant is given by:

$$\alpha = \frac{R}{2L} \tag{14}$$

The quality factor Q, which determines the selectivity of the circuit, is expressed as:

$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 RC} = \frac{\omega_0}{2\alpha} = \frac{1}{R}\sqrt{\frac{L}{C}} \tag{15}$$

In the transient regime, the solution to the differential equation of the RLC circuit depends on the relationship between α and the natural (resonant) frequency ω_0:

- Underdamped regime ($\alpha < \omega_0$):

$$i(t) = e^{-\alpha t}(A\cos(\omega_d t) + B\sin(\omega_d t)) \tag{16}$$

where $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$ is the damped oscillation frequency.

- Critically damped regime ($\alpha = \omega_0$):

$$i(t) = (A + Bt)e^{-\alpha t} \tag{17}$$

- Overdamped regime ($\alpha > \omega_0$):

$$i(t) = Ae^{-(\alpha-\beta)t} + Be^{-(\alpha+\beta)t} \tag{18}$$

with $\beta = \sqrt{\alpha^2 - \omega_0^2}$.

Damping directly influences the bandwidth of the circuit, which is inversely proportional to the quality factor:

$$\Delta\omega = 2\alpha = \frac{\omega_0}{Q} \tag{19}$$

These characteristics are essential in designing filters and communication systems, where controlling damping determines system performance.

5.2. Analytical Interpretation of Poles: Resonance and Damping in RLC Circuits

In the analysis of dynamical systems, the transfer function plays an essential role in describing a system's behavior. This is generally a meromorphic function, i.e., a rational function of the complex variable $s \in \mathbb{C}$, with poles as isolated singularities. The poles and zeros of this function completely determine the dynamic structure of the system, influencing stability, resonant frequencies, and the system's ability to attenuate or amplify certain components of the input signal.

5.3. Transfer Function of an RLC Circuit and the Characteristic Equation

Consider a series RLC circuit, composed of a resistor R, an inductor L, and a capacitor C. Applying the Laplace transform to the differential equation describing the circuit's behavior and assuming zero initial conditions, we obtain the transfer function (e.g., admittance, voltage ratio, etc.). If considering admittance $Y(s) = I(s)/V_{\text{in}}(s)$:

$$H(s) = Y(s) = \frac{I(s)}{V_{\text{in}}(s)} = \frac{1}{Ls+R+\frac{1}{sC}} = \frac{sC}{LCs^2+RCs+1}$$

Or if considering voltage across capacitor $V_C(s)/V_{in}(s)$:

$$H(s) = \frac{V_C(s)}{V_{in}(s)} = \frac{1/sC}{Ls+R+1/sC} = \frac{1}{LCs^2+RCs+1}$$

Let's assume the characteristic equation is related to the denominator:

$$LCs^2 + RCs + 1 = 0 \quad \text{orequivalently} \quad Ls^2 + Rs + \frac{1}{C} = 0$$

Solving this quadratic equation, we obtain the system poles:

$$s_{1,2} = \frac{-R \pm \sqrt{R^2 - 4\frac{L}{C}}}{2L} = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

These poles determine the dynamic behavior of the system:

- If $R^2 > 4\frac{L}{C}$ $((R/2L)^2 > 1/LC)$, the poles are real and negative – overdamped regime.
- If $R^2 = 4\frac{L}{C}$ $((R/2L)^2 = 1/LC)$, the poles are real and equal – critical damping.
- If $R^2 < 4\frac{L}{C}$ $((R/2L)^2 < 1/LC)$, the poles are complex conjugates – underdamped regime, with damped oscillations.

5.4. Poles and Transient Behaviour

In the case where the poles are complex conjugates, $s_{1,2} = -\alpha \pm j\omega_d$, the system's impulse response (related to current or voltage) is typically of the form:

$$\text{response}(t) \propto e^{-\alpha t}(A\cos(\omega_d t) + B\sin(\omega_d t))$$

where:

- $\alpha = \frac{R}{2L}$ is the damping coefficient,
- $\omega_0 = \frac{1}{\sqrt{LC}}$ is the natural (undamped resonant) frequency,
- $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$ is the damped frequency.

This formula highlights the direct link between the position of the poles and the system's behavior over time: the smaller α (poles closer to the imaginary axis), the longer the oscillations persist, indicating weak damping and proximity to resonance.

5.5. Resonance and Quality Factor

Resonance occurs in RLC circuits when the inductive and capacitive reactances cancel each other out, i.e., for AC analysis at frequency ω_0:

$$\omega_0 L = \frac{1}{\omega_0 C} \Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}$$

At this frequency, the impedance of a series RLC circuit is minimal:

$$Z(\omega_0) = R$$

and the current is maximum, according to Ohm's law: $I = \frac{V}{Z}$.

The degree of selectivity and the "sharpness" of the resonance peak is given by the quality factor Q:

$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 RC} = \frac{\omega_0}{2\alpha} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

Interpretation:

• A high Q indicates poles close to the imaginary axis, meaning persistent oscillations and a pronounced resonance.

• A low Q indicates poles deeper in the left half-plane, meaning high damping and broad filtering.

6. Meromorphic Functions and Stability

The transfer function $H(s)$ is a meromorphic function, meaning it is analytic throughout the complex plane, except for a finite number of poles (isolated singularities). These functions naturally arise in the analysis of linear time-invariant (LTI) systems because they allow for:

• Frequency domain analysis (via Laplace or Fourier transforms),

• Locating poles, which correspond to the system's natural modes (eigenvalues),

• Evaluating stability: the system is stable if and only if all poles are in the left half-plane $\Re(s) < 0$.

The meromorphic representation of the transfer function provides a robust mathematical framework for studying:

• Resonance (when the input signal frequency approaches the natural frequency),

• Damping (through the damping coefficient α, the real part of the poles),

• Stability (through the sign of the real part of the poles).

6.1. Correlation between Poles, Resonance, Damping, and Stability

The poles determine the transient response and stability of the system. Their position in the complex plane directly influences:

• Resonance: when the imaginary part is dominant (poles near the $j\omega$ axis), the system exhibits persistent oscillations.

• Damping: when the real part is large (negative, i.e., far left), the system stabilizes quickly with minimal oscillations.

• Stability: if all poles have negative real parts, the system is stable.

Thus, analyzing the poles of the transfer function, a meromorphic function, provides a deep understanding of the dynamic behavior of RLC circuits and dynamical systems in general.

Conclusions

1. Circuit Impedances

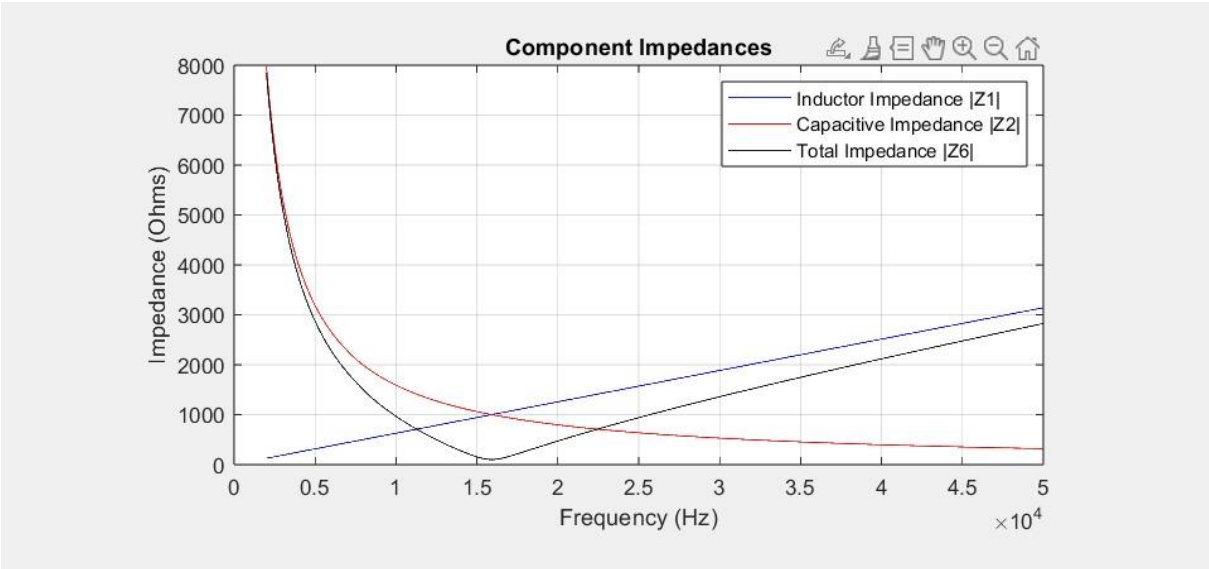

Figure 1: Circuit Impedances

Interpretation of the graph:

The graph highlights the variation of the main impedances of an RLC circuit as a function of frequency. Each curve corresponds to an element or a combination of elements, and their shape reflects the distinct frequency-dependent character of each component. (Assuming Z1 is XL, Z2 is XC, Z6 is total Z based on shape)

• **Inductive Impedance (Reactance):**

$$|X_L(f)| = 2\pi f L$$

increases linearly with frequency f, as the inductor opposes current proportionally to the rate of change of the current. This behavior is typical for coils.

• **Capacitive Impedance (Reactance):**

$$|X_C(f)| = \frac{1}{2\pi f C}$$

decreases hyperbolically with increasing frequency; the capacitor offers higher impedance to slow signals and becomes more transparent at high frequencies.

• **Total Impedance:** Combines all circuit elements (assuming a specific configuration like series L, C with parallel R, Rout):

$$|Z_{total}(f)| = \left|j2\pi fL + \frac{1}{j2\pi fC} + R_{eq}\right| \quad \text{(Example: SeriesL, C, Req)}$$

Assuming Z_6 represents total impedance:

$$|Z_6(f)| = \left|j2\pi fL + \frac{1}{j2\pi fC} + \left(\frac{1}{\frac{1}{R}+\frac{1}{R_{\text{out}}}}\right)\right|$$

This exhibits non-linear behavior, having a local minimum near the resonant frequency.

Resonance:

At the resonant frequency f_0, the inductive and capacitive reactances cancel:

$$2\pi f_0 L = \frac{1}{2\pi f_0 C} \quad \Rightarrow \quad f_0 = \frac{1}{2\pi\sqrt{LC}}$$

At this point, the impedance becomes minimal (if series L, C) or maximal (if parallel L, C), and the circuit allows maximum current flow (series) or minimum current flow (parallel). The graph shows a "V" shape, typical for the magnitude of impedance in a series RLC circuit near resonance.

Meromorphic Analysis:

The total impedance $Z(f)$, viewed as a function of the complex frequency $s = \sigma + j\omega = \sigma + j2\pi f$, can be expressed as:

$$Z(s) = sL + \frac{1}{sC} + R_{\text{eq}}, \quad \text{where} \quad R_{\text{eq}} = \left(\frac{1}{\frac{1}{R}+\frac{1}{R_{\text{out}}}}\right) \quad \text{(Usingthestructurefromthegraphformula)}$$

This is a rational function in s, hence a meromorphic function: holomorphic on the complex plane, except for poles (values where $|Z(s)| \to \infty$).

Interpretation in the complex plane:

Approaching a pole (e.g., where $s \to 0$ or $|s| \to \infty$) corresponds to regions where the impedance increases sharply. The resonant frequency, on the other hand, is a point of equilibrium between these extreme behaviors, often reflecting a cancellation of the imaginary part of the impedance (for $s = j\omega$).

Conclusion:

The graph provides a visual representation of the interaction between reactive and resistive elements. Interpreting this in terms of meromorphic functions allows for rigorous modeling of this phenomenon. Thus, the mathematical analysis of total impedance offers not only information about the circuit's behavior but also a direct application of complex function theory in electrical engineering.

2. Current as a Function of Frequency

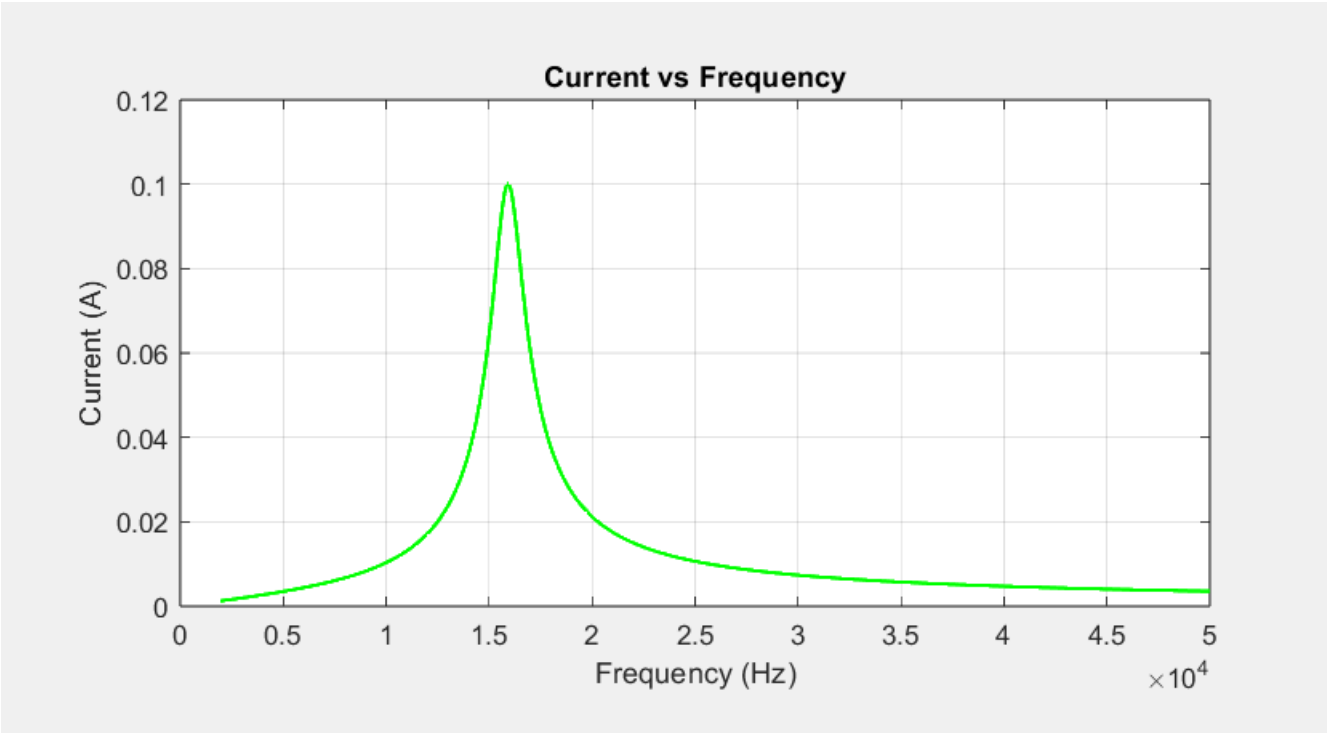

Figure 2: Current as a Function of Frequency

Interpretation of the graph:

The graph illustrates the variation of the current intensity I (assuming I_1 is the total current) through the circuit as a function of the supply signal frequency. A sharp increase in current around a central frequency (the resonant frequency f_0) is observed. This indicates typical selective filtering behavior, where the RLC circuit favors the passage of signals within a narrow frequency range.

Behavior outside resonance:

- At low frequencies, capacitive impedance dominates:

$$Z_C = \frac{1}{j\omega C} \quad \Rightarrow \quad |Z_C| \text{islarge} \quad \text{for} \quad f \ll f_0$$

- At high frequencies, inductive reactance becomes increasingly significant:

$$Z_L = j\omega L \quad \Rightarrow \quad |Z_L| \text{islarge} \quad \text{for} \quad f \gg f_0$$

In both cases (for a series circuit), the total impedance is large, and the current $I(f)$ is reduced.

Resonance:

At the resonant frequency: $f_0 = \frac{1}{2\pi\sqrt{LC}} \quad \Rightarrow \quad \omega_0 L = \frac{1}{\omega_0 C}$

Reactances cancel each other out, and the total impedance becomes purely resistive (minimal for series):

$$Z(f_0) = R_{\text{eq}} \quad \Rightarrow \quad I_{\max} = \frac{V_0}{R_{\text{eq}}}$$

This point corresponds to the peak observed in the graph, where the current reaches its absolute maximum.

Mathematical modeling:

The current intensity is given by the relationship:

$$I(f) = \frac{V_0}{|Z(f)|} = \frac{V_0}{\left|j\omega L + \frac{1}{j\omega C} + R_{\text{eq}}\right|} \quad (\text{where} \omega = 2\pi f)$$

The complex current $I(s) = V_0/Z(s)$ is related to a meromorphic function, characterized by:

• Holomorphism throughout the complex plane except at the poles (where $Z(s) = 0$, if V_0 is constant, or poles of $V_0(s)$ or $1/Z(s)$). The poles of $I(s)$ are typically the poles of $1/Z(s)$.

• A local maximum in magnitude (amplitude peak) near the resonant frequency on the $j\omega$ axis.

Analysis in the complex plane:

The magnitude of the current function $|I(j\omega)|$ has a pronounced maximum around the resonant frequency, behavior that can be associated with the structure of the poles of the transfer function (e.g., admittance $Y(s) = I(s)/V_{in}(s)$). The closer the poles are to the imaginary axis, the narrower and higher the peak, a characteristic expressed by the quality factor Q:

$$Q = \frac{f_0}{f_2 - f_1} = \frac{\omega_0}{\Delta\omega}$$

where f_1 and f_2 are the frequencies at which the current drops to:

$$I = \frac{I_{\max}}{\sqrt{2}}$$

This criterion defines the half-power bandwidth and is essential in characterizing resonance.

Conclusion: Graph 2 emphasizes the circuit's energy efficiency at resonance, where the current is maximum and losses in reactive components (ideally) do not dissipate power over a cycle. Modeling the current's behavior through meromorphic functions provides a robust analytical framework for understanding and designing selective circuits, with applications in filtering, communications, and power electronics.

3 Voltages Across Components

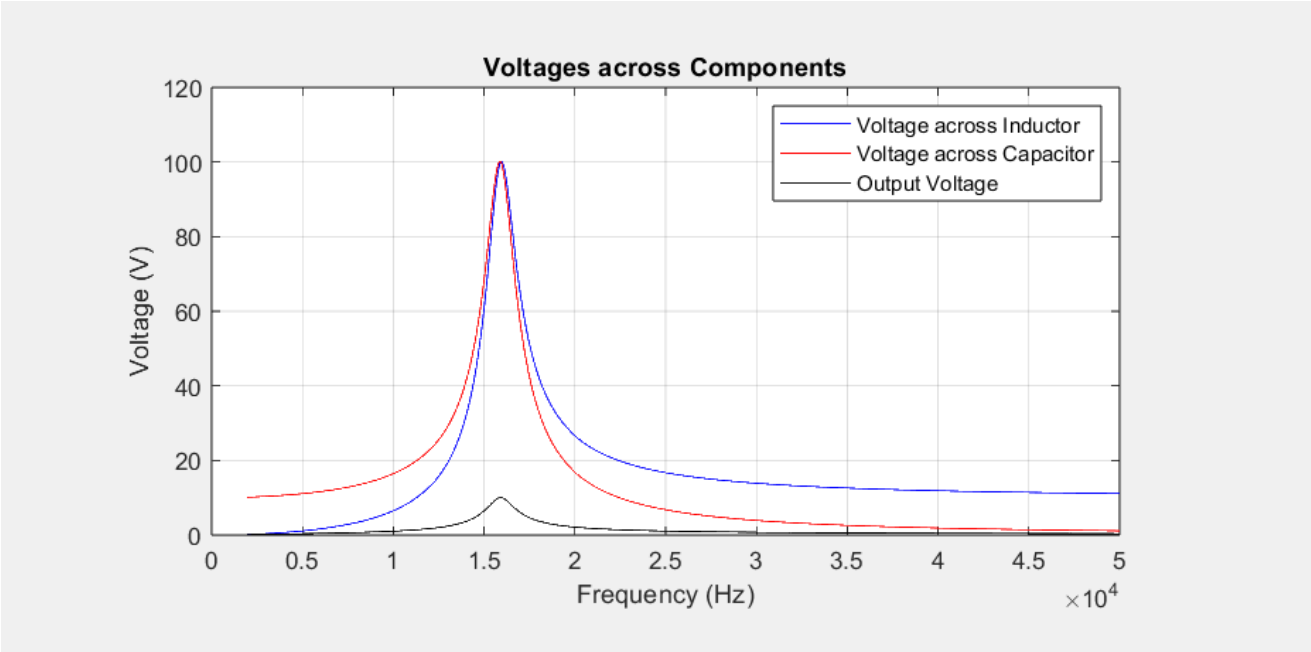

Figure 3: Voltage Across Components

Interpretation of the graph:

The graph highlights the evolution of voltages across the inductor (V_L), capacitor (V_C), and at the output (V_{OUT}) as a function of the input signal frequency. A significant amplification of the voltages across the inductor and capacitor is noticeable around the resonant frequency f_0, where the overvoltage effect specific to RLC circuits occurs (especially in series circuits).

Voltage across the inductor V_L:

$$Z_L = j\omega L \quad \Rightarrow \quad V_L(\omega) = I(\omega) \cdot j\omega L$$

The voltage across the coil generally increases with frequency (X_L increases), but reaches a maximum near the resonant regime, where the current $I(\omega)$ is also maximal. This peak results from the combination of increasing inductive reactance and the intense current at resonance.

Voltage across the capacitor V_C:

$$Z_C = \frac{1}{j\omega C} \quad \Rightarrow \quad V_C(\omega) = I(\omega) \cdot \frac{1}{j\omega C}$$

Complementarily, the voltage across the capacitor is large at low frequencies and decreases as $1/\omega$, but in the resonant regime, due to the large current value, it also reaches a maximum.

Output Voltage V_{OUT}:

$$V_{\text{OUT}} = I(\omega) R_{\text{OUT}} \quad (\text{Example: if output is across } R_{OUT})$$

Near resonance, V_L and V_C are (nearly) 180 degrees out of phase and can be much larger than the input voltage, but their sum is relatively small. If V_{OUT} is taken across the resistor R_{eq}, its shape will mirror the current shape. The graph seems to show V_{OUT} peaking at resonance, consistent with voltage across R in a series RLC. The graph shows U_L and U_C peaking above the input level near resonance, while U_{OUT} also peaks but potentially less dramatically depending on Q and where it's measured.

Resonance and Overvoltages:

At the resonant frequency:

$$\omega_0 L = \frac{1}{\omega_0 C} \quad \Rightarrow \quad Z_{\text{total}}(\omega_0) = R_{\text{eq}} \quad \text{(forseriesresonance)}$$

The current becomes maximum, and the voltages across the inductor and capacitor can exceed the supply voltage ($|V_L| = Q|V_{in}|, |V_C| = Q|V_{in}|$ at resonance for series RLC). These peaks are a consequence of energy storage in reactive components and energy exchange between them.

Meromorphic Modeling:

Each voltage can be expressed as the product of the current and the local impedance (or as a voltage division from the input):

$$V_L(s) = I(s) \cdot sL = V_{in}(s)\frac{sL}{Z(s)}, \quad V_C(s) = I(s) \cdot \frac{1}{sC} = V_{in}(s)\frac{1/sC}{Z(s)}$$

where (for the assumed series L, C, Req structure):

$$Z(s) = R_{\text{eq}} + sL + \frac{1}{sC}$$

The functions $V_L(s)/V_{in}(s)$ and $V_C(s)/V_{in}(s)$ are rational functions in s, with poles determined by the roots of the denominator $Z(s)$, hence they are meromorphic functions (transfer functions).

Interpretation in the complex plane:

Near the resonant frequency ($s \approx \pm j\omega_0$), the behavior of these functions reflects the proximity to the complex conjugate poles of the system, leading to a local amplification of the magnitude (overvoltage). The width of the frequency range where these overvoltages occur is regulated by the quality factor Q, which depends on the ratio of stored energy to dissipated energy per cycle.

Conclusion:

Graph 3 highlights the role of reactive components in shaping the voltages within the circuit. Local overvoltages, behavior at frequency limits, and amplification are all fundamental characteristics governed by the meromorphic functions describing the circuit's frequency

response. These properties are essential in designing filters, oscillators, and efficient energy transfer systems.

4 Signals in the Time Domain

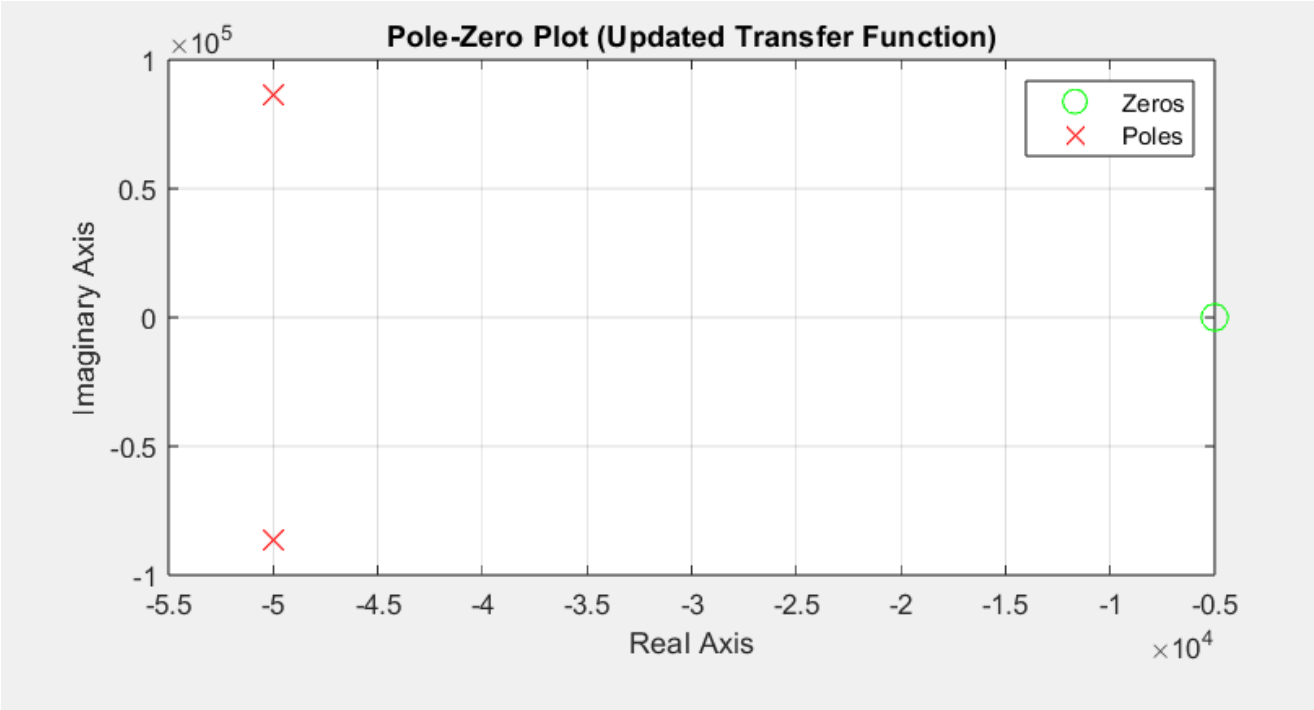

Figure 4: Graphical Representation of the Poles and Zeros of the Transfer Function

Interpretation of the graph :

This graph illustrates the locations of the poles (marked by 'X') and zeros (marked by 'O') of the system's transfer function $H(s)$ in the complex s-plane ($s = \sigma + j\omega$). This plot provides crucial insights into the system's dynamic behavior, including stability, response speed, and frequency characteristics.

Transient and Steady-State Behavior Connection:

The time-domain response of the system (like $V_{OUT}(t)$ for an input $V_{IN}(t)$) is directly related to the pole locations. For instance, a pair of complex conjugate poles $s = -\alpha \pm j\omega_d$ corresponds to an oscillatory component in the transient response:

$$\text{Component}(t) \propto e^{-\alpha t}\cos(\omega_d t + \phi)$$

where: • $\alpha = -\text{Re}(s)$ represents the damping rate (related to distance from the $j\omega$ axis). Larger α means faster decay.

• $\omega_d = \text{Im}(s)$ is the damped oscillation frequency.

• Stability requires all poles to be in the left half-plane ($\text{Re}(s) < 0$, i.e., $\alpha > 0$).

The steady-state response to a sinusoidal input $V_{IN}(t) = V_0\cos(\omega t)$ is given by $V_{OUT}(t) = V_0|H(j\omega)|\cos(\omega t + \angle H(j\omega))$. The magnitude $|H(j\omega)|$ and phase $\angle H(j\omega)$ are determined by the distances and angles from the point $j\omega$ on the imaginary axis to the poles and zeros.

Meromorphic Functions and System Response:

The transfer function $H(s)$ is inherently a meromorphic function (typically rational for lumped LTI systems):

$$H(s) = \frac{V_{\mathrm{OUT}}(s)}{V_{\mathrm{IN}}(s)} = K\frac{(s-z_1)(s-z_2)\ldots}{(s-p_1)(s-p_2)\ldots}$$

where z_i are the zeros and p_i are the poles.

• **Poles (X):** Determine the natural modes of the system.

- Stability: All poles must be in the Left Half-Plane (LHP).

- Response Speed/Damping: Distance from the $j\omega$ axis ($-\mathrm{Re}(p_i)$).

- Oscillation Frequency: Imaginary part ($\mathrm{Im}(p_i)$). Poles close to the $j\omega$ axis indicate low damping and potential resonance near $\omega = |\mathrm{Im}(p_i)|$.

• **Zeros (O):** Influence the amplitude and phase of the response at specific frequencies.

- Frequencies where the response magnitude is zero (if $s = j\omega$ is a zero).

- Shape the frequency response curve $|H(j\omega)|$. Zeros near the $j\omega$ axis cause dips in the magnitude response.

Physical Relevance:

The pole-zero plot graphically explains the system's behavior observed in frequency response (like Graphs 1-3) and time-domain signals.

• Resonance corresponds to poles being close to the $j\omega$ axis. The closer the poles, the higher the Q factor and the sharper the resonance peak in $|H(j\omega)|$.

• Damping is directly related to the real part of the poles. Poles far into the LHP lead to heavily damped, non-oscillatory responses.

• Stability is immediately assessed by checking if all poles lie in the LHP.

The signals shown in time-domain plots (like potentially the previous graph, if it showed time signals) for frequencies near resonance (f_0) would exhibit larger amplitudes and less decay compared to frequencies far from resonance, directly reflecting the influence of the nearby poles. Frequencies f_1, f_2 (half-power points) correspond to specific points on the magnitude response $|H(j\omega)|$ related to the pole locations.

Final Conclusion: This work rigorously highlights the essential role that meromorphic functions play in the analysis of dynamical systems and electrical circuits, offering a solid mathematical framework for understanding phenomena such as resonance, damping, and stability. By their nature, meromorphic functions (holomorphic functions except for a finite number of poles) allow for the precise description of physical system behavior through their analytical structure and the distribution of their singularities.

Applying these functions to the analysis of RLC circuits prooved that the position of poles in the complex plane determines not only the transient response and steady-state regime of the system but also its ability to filter, amplify, or attenuate signals. Thus, we saw how a meromorphic transfer function, expressed as a ratio of polynomials in the complex variable s, faithfully reflects a system's behavior towards different input frequencies. Resonance emerges as a natural phenomenon of the pole structure, and damping is dictated by their real part.

More than just a computational tool, the transfer function becomes an analytical map synthesizing both the physical properties (such as natural frequencies, damping, or quality factor) and the stability characteristics of a system. Through the analyzed graphs, it was observed how meromorphic functions govern the behavior of impedance, current, and voltages in the frequency domain, as well as the shape of signals in the time domain. All these results reflect the power of complex function theory in modeling and deeply understanding engineering phenomena.

In conclusion, meromorphic functions represent not only a theoretical concept of mathematical interest but also an essential tool for engineering and physics, with direct applications in:

- Design of electrical systems
- Analysis of dynamic stability
- Optimization of circuit response

The connection between mathematics and physics is thus strengthened by this symbiosis between the formalism of complex analysis and the reality of the dynamic behaviors of technical systems.

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